MTH 451/551, Fall 2026
Quiz #1, Tue., Oct. 6, 2026

Solutions

1. Let $f(x) = \sin(x)$, and let $c = 0$. Recall the nth order Taylor polynomial of f at c is

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

(a) Find $p_1(x)$, the first order Taylor polynomial of f at c. What is its degree?

(b) Find $p_2(x)$, the second order Taylor polynomial of f at c. What is its degree?

Solution. We compute sin and its derivatives at $c = 0$,

$$\sin(0) = 0, \qquad \sin'(0) = \cos(0) = 1, \qquad \sin''(0) = -\sin(0) = 0, \qquad \sin^{(3)}(0) = -\cos(0) = -1,$$

and then write

$$p_1(x) = \sin(0) + \sin'(0)x = x,$$
$$p_2(x) = \sin(0) + \sin'(0)x + \tfrac{1}{2}\sin''(0)x^2 = x.$$

The degree of both polynomials is 1.

2. Consider the forward difference approximation to the derivative

$$f'(c) \approx \frac{f(c+h) - f(c)}{h} =: F.$$

(a) In general, if h is halved, what is expected to happen to the error $|f'(c) - F|$?

(b) What happens in the specific case of $f(x) = \sin(x)$ and $c = 0$? (Hint: use Taylor's Theorem with p_2 from question 1).

Solution. The error in the forward difference approximation to the derivative is given by the expression

$$f'(c) - \frac{f(c+h) - f(c)}{h} = -\frac{f''(\xi)}{2}h,$$

for some ξ between c and $c + h$. The right-hand side scales like h, so if h is halved, we expect the error to be roughly halved as well. The error will not generally be *exactly* halved, because we evaluate f'' at a different ξ.

In the specific case of $f(x) = \sin(x)$ and $c = 0$, note that $f''(\xi) = -\sin(\xi)$, and for ξ close to 0, $\sin(\xi) \approx \xi$, so this suggests the error may actually be smaller than predicted above. We can make this rigorous using $p_2(x)$. Taylor's Theorem with h gives

$$\sin(h) = p_2(h) + \frac{\sin'''(\xi)}{3!}h^3 = h - \frac{\cos(\xi)}{6}h^3.$$

The forward difference quotient is

$$\frac{f(c+h) - f(c)}{h} = \frac{\sin(h) - \sin(0)}{h} = \frac{h - \frac{1}{6}\cos(\xi)h^3}{h}$$
$$= 1 - \frac{1}{6}\cos(\xi)h^2 = \sin'(0) - \frac{1}{6}\cos(\xi)h^2,$$

so, because the first and second Taylor polynomials are the same, we get an extra cancellation, and in this particular case, if we halve h, we expect the error to decrease by a factor of about 4.